ReconIQ

Research-Informed Trade Reconciliation and Exception Screening

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1. Problem and real-world context

Area, user and task

ReconIQ looks at a routine post-trade task in investment operations. An analyst compares the firm's order-management or portfolio-accounting record with the broker's or custodian's confirmation, links the two records, and investigates any differences. The fields in scope are security ID, trade date, buy / sell side, quantity, price, currency, settlement date and account.

Problem statement: who, what, how often and cost

The work starts when a trade or a batch of confirmations arrives. Timing varies by firm: checks may happen intraday or at end of day, depending on data feeds and settlement cut-offs. This prototype uses two CSV files to demonstrate a batch run; it does not assume one standard industry schedule. The main cost to measure is analyst time spent finding breaks, checking source records, correcting links and escalating unresolved items. I did not have data from an operations team, so I have not attached a cost or savings estimate to the problem.

A firm can estimate its current workload from four inputs: trades per run, the share that need manual review, average minutes per review and the analyst's loaded hourly cost. Multiply the number of reviewed trades by the time per review, then by the hourly cost. Measure rework and late-settlement effects separately. The worksheet in docs/business_case_discovery.md lists the information to collect.

Why the problem matters and why now

A false match can hide a real difference between two trades. A loose similarity rule can create the opposite problem by sending an analyst to investigate the wrong record. The proposed first step is deliberately modest: read approved CSV exports and confirmation notes, show the supporting evidence, and leave decisions to the analyst. ML is used only to help screen narrative notes; structured trade fields are checked with explicit rules.

Why this fits Tresata

- Tresata's note on Medallion Architecture describes staging, profiling, standardising, record linkage and aggregation. It also names trade reconciliation as a use case [1].
- ISDA's *Portfolio Reconciliation in Practice* discusses operational issues such as inconsistent account names, wrong-entity booking and incorrect settlement instructions [2].
- TCS describes OTC reconciliation exceptions including incorrect account data, UTI mismatches, timing differences and duplicate bookings [3]. I use this vendor paper as an example of the workflow, not as an independent measure of market-wide performance.
- Public practitioner discussions were also reviewed, but only as anecdotal context. They are not evidence of how common a break is or how much time it costs.

A useful design observation

The workflow contains three different questions: is a value present in the confirmation, does a note describe an exception, and do the two trade records agree? The answers can differ. A note may contain an extractable quantity and describe a quantity break, while the trade pair is still uncertain because the candidate link is ambiguous. ReconIQ keeps extraction, text screening and trade comparison separate so that one successful step is not mistaken for proof of a match.

Scope, jurisdiction and exclusions

The demo covers two-way, one-to-one comparisons using generic equity and simple fixed-income examples. It is not a tax, legal or regulatory rules engine and contains no jurisdiction-specific legal conclusion. A real pilot would need to select its markets and instruments, confirm identifier standards, settlement calendars, time zones and broker formats, and agree effective-dated tolerances. Partial fills, allocations, novations, lifecycle events, multi-leg OTC products, corporate actions, cash / position reconciliation, fees, FX conversion, valuation, margin, collateral, settlement execution and ledger write-back are out of scope. Only synthetic or explicitly approved public data is used .

Why I selected trade reconciliation

I initially explored finance workflows where data coming from different systems has to be compared and exceptions have to be investigated. While reading Tresata's *The Accidental Medallion Architecture*, I noticed that trade reconciliation was specifically mentioned as a use case for the data-processing approach described in the note, particularly staging, profiling, standardisation, semantic joins and record linkage. This made me look more closely at how reconciliation works in practice. I also explored public discussions in communities such as r/FinancialCareers, r/CFA and r/Accounting. A discussion in r/FinancialCareers described the operational work involved in investigating trade and position breaks, checking trades against CTM and ensuring that they reach the custodian. What caught my attention in r/CFA was a more specific issue: different representations of fixed-income securities between internal systems and custodians, along with settlement-date assumptions such as T+1 versus T+2. Another discussion in r/Accounting described the difficulty of comparing thousands of transactions when amounts are split, descriptions differ, and duplicates or missing records are present. These discussions helped me understand that reconciliation is not always a straightforward row-by-row comparison. A record may represent the same underlying trade even when some fields are formatted differently, while a genuine difference in quantity, security identity or settlement information should not be ignored just to force a match. That led me to choose trade-level reconciliation as the problem for ReconIQ. I wanted to explore a workflow that combines structured field comparisons, identification of potential exceptions and screening of free-text notes, while leaving ambiguous cases for an analyst to review. I used the Reddit discussions as qualitative inputs to problem discovery, rather than treating them as statistical evidence of industry-wide performance.

The practitioner discussions also influenced this decision. In particular, the r/CFA discussion about T+1 and T+2 settlement assumptions made me realise that a numerical or date difference cannot always be interpreted without considering the underlying business rule. Similarly, the discussion about different fixed-income security representations showed why forcing a match based only on apparent numerical similarity could be risky. This reinforced my decision to keep structured comparisons explainable and to use an explicit review path for cases where the system cannot confidently establish a match.

2. Research foundation and method transfer

I studied and referred these two papers that I found in my research to find the optimal problem statement for this project scouring throught the various parts of the internet.

Paper A - Han, Wang and Huang (2026), IEEE Access

Han, Wang and Huang study IFRS 16 right-of-use lease-contract intelligence, not trade reconciliation. Their ROU-TransQA-v2 system separates the question of whether evidence exists from the task of locating the supporting text. It then passes the evidence to a separate indicator-mapping layer. The useful idea for ReconIQ is this separation, not the paper's lease rules.

In the paper's independent CUAD test set, 68.20% of question instances were unanswerable. Across five runs, HasAns F1 was 21.54 +/- 0.38 and NoAns F1 was 99.34 +/- 0.12. This is a useful warning: a system can be strong at saying that evidence is absent and still struggle to extract evidence when it is present. ReconIQ therefore keeps NOT_FOUND and AMBIGUOUS as valid outcomes and stores the source quote for each extracted value [4].

Paper B - Lippi et al. (2019), CLAUDETTE

CLAUDETTE studies potentially unfair clauses in online Terms of Service, not financial exceptions. It separates sentence-level detection from the later task of assigning one or more categories. Its corpus contains 50 contracts and 12,011 sentences. Three annotators developed the guidelines, and a separate ten-contract set achieved Cohen's kappa of 0.871. A sentence could receive more than one label [5].

The authors compare lexical SVMs, category-specific SVMs, tree kernels, CNNs, LSTMs, SVM-HMMs and an ensemble. Their category-specific SVM combination, C2, achieved P/R/F1 of 0.798 / 0.782 / 0.781. The C8 ensemble achieved 0.826 / 0.797 / 0.805 on leave-one-document-out evaluation, but the paper's web service used C2 because it was faster [5].

The paper's error analysis shows why a keyword-only screen can fail: a sentence may mention a topic without expressing the issue being screened for, and context can change its meaning. The authors discuss context modelling and targeted post-processing. ReconIQ follows the general lesson by showing adjacent sentences and exposing the raw detector flag separately from its post-processed result [5].

Method choices adapted from CLAUDETTE

Paper method	ReconIQ adaptation
Binary sentence detection, followed by categorisation	Detect potential trade-exception text, then categorise the issue
Eight category-specific SVMs; multi-label outputs	Eight one-vs-rest trade-break SVM labels; a note can have multiple labels
Word / POS TF-IDF and n-grams	Word and character n-gram TF-IDF with Linear SVM
Document / sentence context and error analysis	Adjacent sentence window, source offsets, FP/FN review and narrow post-processing
Guidelines, independent annotation and agreement	Synthetic label manual now; expert double-annotation and agreement are future validation

ReconIQ does not reuse the ToS corpus, legal category names, fairness ratings, SVM-HMM, C8 ensemble or CLAUDETTE's reported scores. Its classifier is a small synthetic TF-IDF / Linear SVM prototype. It is not a reproduction of C2 or C8.

3. Solution architecture and decision authority

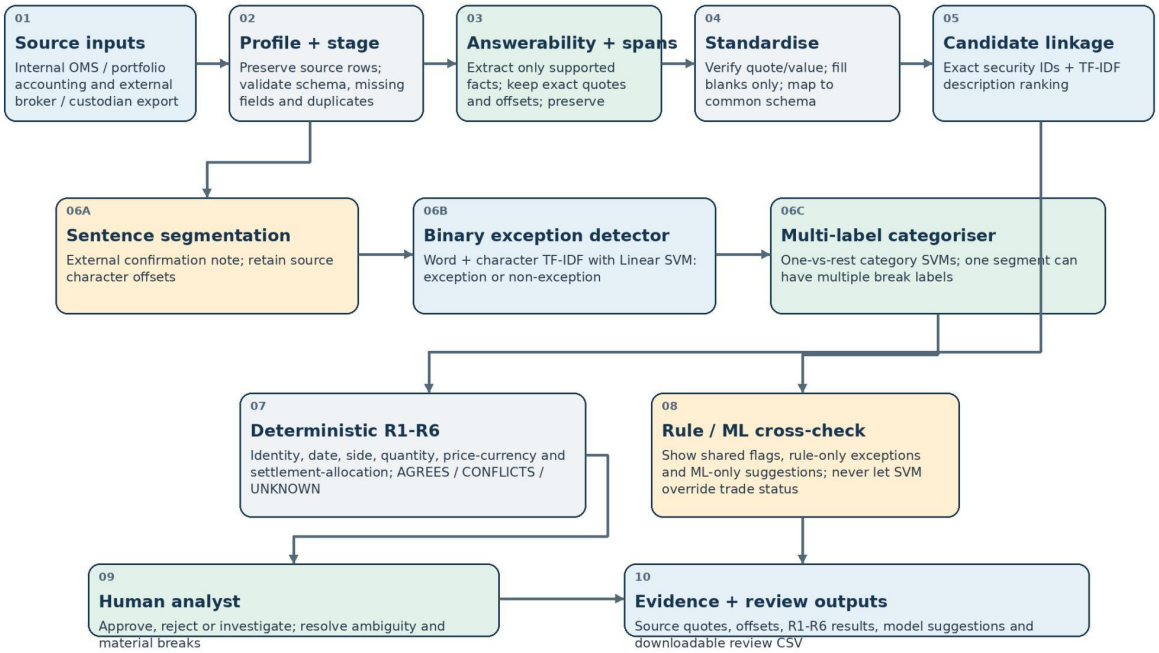


Figure 1. Evidence extraction, text screening and deterministic reconciliation are separate stages; the human analyst owns the final review action.

How the prototype works

1. Preserve internal and external CSV rows; profile schema, missing fields and duplicate source IDs.
2. Extract explicitly labelled values from external confirmation notes. Keep ANSWERABLE / NOT_FOUND / AMBIGUOUS, the exact quote and offsets; fill a blank structured field only from a verified source value.
3. Generate candidate pairs with exact security IDs and character n-gram TF-IDF description similarity.
4. Screen note segments with a binary Linear SVM; if flagged, suggest one or more categories with one-vs-rest SVMs. Retain adjacent context and offsets.
5. Run deterministic R1-R6 comparisons; expose rule / ML disagreements; route unresolved cases to an analyst and export the result.

R1-R6 deterministic evidence map

Dimension	Compared evidence	Possible state
R1 Identity	Exact security ID; description is candidate support only	AGREES / CONFLICTS / UNKNOWN
R2 Trade date	Execution date	AGREES / CONFLICTS / UNKNOWN
R3 Side	BUY / SELL	AGREES / CONFLICTS / UNKNOWN
R4 Quantity	Units / face value / notional	AGREES / CONFLICTS / UNKNOWN
R5 Price-currency	Price tolerance and currency	AGREES / CONFLICTS / UNKNOWN
R6 Settlement-allocation	Settlement date, account, counterparty	AGREES / CONFLICTS / UNKNOWN

A pair is labelled MATCHED - EXACT only when the security identifier is exact, the candidate is unique and mutual-best, there is no duplicate or competing link, and all six dimensions agree. UNKNOWN is not agreement. The SVM margin is a ranking score, not a probability, and it cannot change the deterministic trade status.

Why use ML at all?

Rules are the right tool for fields such as identifiers, dates, side, quantity, currency and price tolerance. The text model has a narrower possible role: it can flag notes where the wording varies or one sentence describes more than one break. That may help an analyst find what to review, but the current synthetic test does not show that ML beats a well-maintained rules-only screen on real broker notes. The next test is to compare rules-only and rules-plus-ML on the same approved sample. If ML does not improve detection or review time without creating too many false alerts, it should be removed.

The design avoids four extremes. An exact join alone can miss records when identifiers or descriptions differ. Fuzzy similarity cannot prove that two trades are the same. An LLM-only design is unnecessary for numeric field checks and is harder to audit. A multi-agent or large-ensemble design is not justified by a small synthetic dataset. The current version therefore uses deterministic checks for trade fields and a small SVM only for note screening.

4. Prototype evidence, risks and next steps

Current synthetic prototype evidence

Layer	Current fixture result	Interpretation
Answerability parser	23 / 23 expected states and values; 11 / 11 answerable quotes verified	Hand-authored parser fixture, not an ML benchmark
Exception classifier	198-row synthetic pool (156 positive / 42 negative); 30 grouped validation rows; 50 held-out tests	Phrase-family fixture, not an independent finance corpus
Binary detection	Raw SVM P/R/F1 = 0.9756 / 1.0000 / 0.9877; post-processed = 1.0000 / 1.0000 / 1.0000	Small synthetic held-out sample; not production performance
Multi-label categories	Macro F1 = 0.9218; exact label-set accuracy = 0.90; five exact-set errors	Co-occurring issue labels remain imperfect
Trade reconciliation	13 internal + 13 external; 3 exact, 9 review, 1 unmatched per side, 1 duplicate group	Controlled synthetic trade examples

Selected classifier examples

Case	Synthetic scenario	Expected	Predicted	Result
EV0001	Security identifier mismatch	SECURITY_IDENTITY	SECURITY_IDENTITY	Pass
EV0005	Execution date mismatch	TRADE_DATE	TRADE_DATE	Pass
EV0009	BUY / SELL side mismatch	SIDE	SIDE	Pass
EV0013	Quantity mismatch	QUANTITY	QUANTITY	Pass
EV0017	Price mismatch	PRICE_CURRENCY	PRICE_CURRENCY	Pass
EM0001	Quantity + currency breaks	QUANTITY + PRICE_CURRENCY	QUANTITY + PRICE_CURRENCY	Pass
EM0002	Account + settlement breaks	ACCOUNT_ALLOCATION + SETTLEMENT_DATE	ACCOUNT_ALLOCATION + SETTLEMENT_DATE	Pass
EM0003	Duplicate record + side break	DUPLICATE_OR_MISSING_LINK + SIDE	SIDE	Fail: missed label
EM0004	Missing link + security mismatch	DUPLICATE_OR_MISSING_LINK + SECURITY_IDENTITY	No labels	Fail: missed both
EM0005	Security ID + trade-date breaks	SECURITY_IDENTITY + TRADE_DATE	TRADE_DATE + PRICE_CURRENCY	Fail: missed / extra

Five of the 50 held-out examples have an incorrect exact label set. The table shows three of those errors; EM0006 and EM0007 are in the full evaluation file. The difference matters: the detector can recognise that a note contains an exception while the category model still misses or adds a label. For that reason, category suggestions remain review items rather than decisions.

Risks, controls and ownership

Risk	Control / limitation
False match / duplicate	Exact unique R1-R6 gate; duplicates and competing links go to review
False text label or omission	Show raw flag, suppressed labels, source text, context and category metrics
Missing / ambiguous evidence	Answerability states and UNKNOWN; no inferred fill
Outdated market rules	Prototype has no market calendar / counterparty rule registry; pilot needs effective-dated, version-controlled configuration and an owner
Confidentiality / liability	Synthetic data only; no live connectivity, settlement or journal posting; authorised human remains accountable

The auditable outputs are the source IDs, extracted quotes and offsets, normalised fields and R1-R6 comparisons. The SVM outputs are suggestions: candidate ranking and possible narrative categories. The margin is not a probability. A real implementation would need controlled, effective-dated mappings, tolerances, calendars and counterparty rules, along with model versioning. Those governance features are not built into this prototype.

The main weakness is the data. The classifier has not been trained or tested on an independently collected, expert-labelled set of broker notes. Its holdout is synthetic, so new counterparties, abbreviations, instruments and lifecycle language may cause errors. It also sees adjacent sentences, not the full confirmation document.

Checks needed before a pilot

First, speak with an operations owner and measure trade volume, break rate, handling time, rework, escalations and loaded labour cost. Then compare manual review, rules-only and hybrid ML on the same approved sample. The sample should be labelled independently by at least two reviewers, with disagreements recorded before adjudication. Testing should include unseen counterparties and instruments. Access controls, source hashes, audit logs, effective-dated rule versions and model monitoring would also be needed before live use.

References

[1] Tresata Tech Notes, “The Accidental Medallion Architecture,” 8 February 2023. https://technotes.tresata.com/medallion-architecture/

[2] ISDA, *Portfolio Reconciliation in Practice*. https://www.isda.org/a/crpTE/Portfolio-Reconciliation-In-Practice.pdf

[3] TCS, “Automated Reconciliation Process for Over-the-counter Derivatives.” https://www.tcs.com/what-we-do/industries/capital-markets/white-paper/automated-reconciliation-process-otc-markets

[4] Han, Y., Wang, Y. and Huang, L. (2026), “An Answerability-Aware Contract Intelligence Framework for Right-of-Use Asset Management With IFRS 16-Oriented Indicator Mapping and Multidimensional Analytics,” *IEEE Access*. DOI: 10.1109/ACCESS.2026.3740089.

[5] Lippi, M. et al. (2019), “CLAUDETTE: an automated detector of potentially unfair clauses in online terms of service,” *Artificial Intelligence and Law*, 27, 117–139. DOI: 10.1007/s10506-019-09243-2

[6] Ops role in equities, r/FinancialCareers, 25 August 2021. Reddit discussion[https://www.reddit.com/r/FinancialCareers/comments/pbcjus]

[7] How do you handle Trade/Position Reconciliation?, r/CFA, 19 March 2021. Reddit discussion[https://www.reddit.com/r/CFA/comments/m8iio1]

[8] How do you compare 5000 transactions between two files?, r/Accounting, 7 September 2026. Reddit discussion[https://www.reddit.com/r/Accounting/comments/1w9mk32]

All prototype trades and training / evaluation examples are synthetic. No market-wide error rate, operating cost or savings figure is claimed.

Appendix: What I Added Beyond AI

Where AI helped

I used ChatGPT and Claude during this assignment to explore the reconciliation workflow, compare the two supplied papers, sketch the architecture, draft and revise code, create synthetic examples, and help interpret test results.

The assistance contributed substantially to the initial implementation and to the wording of the documents. I am not presenting the first version of the code or brief as work completed without AI assistance.

Checks and decisions retained

- The legal ToS examples and legal labels from CLAUDETTE are not used as trade data. They describe a different task and cannot serve as finance labels.
- The research papers' scores are reported only as findings from those papers. ReconIQ's scores come from its own small synthetic fixtures and are not compared as if the datasets were equivalent.
- The SVM is restricted to note screening and category suggestions. Deterministic R1-R6 checks remain the authority for the structured trade comparison.
- A note can describe more than one break, so the categories are multi-label. A prediction is still only a suggestion because the current tests include category-level errors.
- No cost saving or market-wide exception rate is claimed because no target firm's operating data was supplied.

What came from my own work

- I selected trade reconciliation because I wanted to work on a finance problem where data quality has a direct impact on day-to-day operations. While going through Tresata's The Accidental Medallion Architecture, the part that caught my attention was the sequence of staging and profiling data, bringing it into a unified schema, and then using semantic joins and record linkage before aggregation. The note also specifically mentions trade reconciliation as a potential use case.
  I found this interesting because an internal trading system and an external broker or custodian may represent the same trade differently. Even small differences in security identifiers, quantities, account details or settlement information can leave an analyst with an exception to investigate. So, rather than treating reconciliation as simply matching two spreadsheets, I wanted to explore how data standardisation, record linkage and exception screening could work together in one workflow.
- One decision I made was to keep the comparison of structured trade fields rule-based instead of asking an LLM to make the entire matching decision. I considered a single LLM-based approach that would read both files and decide whether two records matched. However, I felt that comparisons involving fields such as trade date, side, quantity, price and currency should give consistent and explainable results. I therefore kept these checks in a deterministic rules layer, with outcomes such as AGREES, CONFLICTS and UNKNOWN. I kept the machine-learning component for screening free-text confirmation notes, where the wording can vary and multiple exception types may appear in the same note. This separation also ensures that a model suggestion cannot override a structured-field conflict or make the final decision on a trade. The analyst remains responsible for reviewing and resolving exceptions.
- One case I looked at was EM0004. The note says: "A missing counterparty confirmation also contains a different security code." The binary detection stage correctly flagged the note as an exception, but the classification stage returned no exception labels. The expected labels were DUPLICATE_OR_MISSING_LINK and SECURITY_IDENTITY. I would not consider this a successful classification. Although the system recognised that the note needed attention, it failed to explain what the possible issues were. I decided not to change the classification thresholds just to make this one example pass, because changing them based on a single synthetic case could introduce false positives elsewhere. Instead, I recorded it as a category-level failure and identified it as an area for improvement. I would add more varied examples involving missing confirmations and security-identifier mismatches, then test those changes on a separate set of examples before accepting them..

A text classifier should not approve a trade. It may highlight a note that appears to describe a quantity break, but the trade records and the approved comparison rules must establish whether the quantities actually differ. When evidence is missing or ambiguous, the result should stay UNKNOWN and go to an analyst. This lets the model help with triage without giving it authority over financial records.